



University of Vavuniya

Second Examination in Information Technology - 2022

First Semester - January/February 2024

IT2114 Data Structures (Practical)

Answer all Questions

Time Allowed : Three hours

Instructions:

- Write your programs in Java.
- Create a new folder with your index no (eg. IT20001) on your PC's desktop and save all files in the folder.
- 1. Write a program to search an element in an array. If an element is found, print the *index* of that element, and if not, print "no such element found in the array".

Verify your program to search the elements 8 and 9 in the array of elements [0, 8, 10, 12, 23, 31, 43].

[10%]

- 2. Write a program to verify whether a given word is a *palindrome*.

[15%]

- 3. Write a program to sort an array elements in an ascending order using *Merge sort* method.

Verify your program for the elements of an array [12, 31, 25, 8, 32, 17, 40, 42].

[20%]

06.2.2024 (5.1.2024)



Stack

IT Stack

4. (a) Write a program to define a *Queue data structure* using an array. [10%]

(b) Write code segments to perform each of the following in your program:

- i. Create a Queue with the four elements [Apple, Banana, Cherry, Date].
- ii. Display the size of the Queue.
- iii. Display the first element in the Queue.
- iv. Display the last element in the Queue.
- v. Check whether the Queue is empty.
- vi. Delete an element from the Queue and print the updated Queue.
- vii. Add the element Orange into the Queue and print the updated Queue. [15%]

5. (a) Write a *Node* class to represent a *Node data structure* that contains a data item and one or more links where a link is a reference to a Node. [10%]

(b) Write a *LinkedList* program to have four nodes linked together to form the following *LinkedList* and print it:

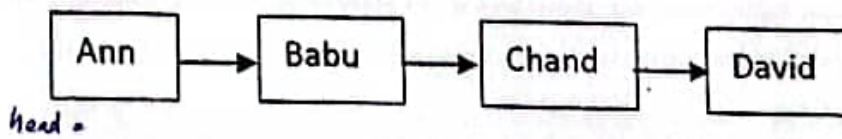


Figure 1: LinkedList

(c) Write each of the following methods in your program:

- i. `addFirst(data)` : To add a node at the first
- ii. `addAfter(Node, data)` : To add a data after a specific node
- iii. `removeFirst()` : To remove the first node
- iv. `removeAfter(Node)` : To remove the node after a specific node [10%]

